

Problem Statement

Title

**IoT Smart Waste Management System
using Raspberry Pi Pico W (Simulation)**

Problem Statement

Municipal waste management faces challenges due to overflowing dustbins, delayed waste collection, and manual monitoring. Overflowing bins create unhygienic conditions, unpleasant odors, and environmental pollution.

The objective of this project is to develop a **simulated IoT Smart Waste Management System** using **Raspberry Pi Pico W** and an **HC-SR04 Ultrasonic Sensor** in the **Wokwi simulation environment**. The system measures the garbage level in real time and transmits the data to an IoT cloud dashboard through Wi-Fi. When the garbage level exceeds a predefined threshold (80%), the system indicates that the dustbin is full.

The project demonstrates how Industrial Internet of Things (IIoT) technology can improve waste collection efficiency and support smart city initiatives.